Electromotive Force

Exam code: 0625 & 0972

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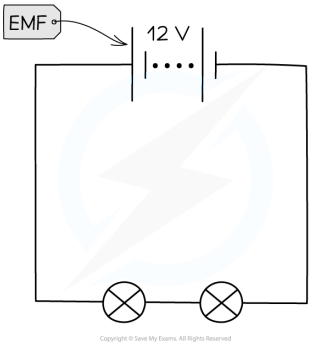
Updated on 30 September 2024

Electromotive force

- Electromotive force (e.m.f.) is the name given to the potential difference of the power source in a circuit
- It is defined as

The electrical work done by a source in moving a unit charge around a complete circuit

- Electromotive force (e.m.f.) is measured in volts (V)



The e.m.f. is the voltage supplied by a power supply: 12 V in the above case

Calculating electromotive force

Extended tier only

- The definition of e.m.f. can also be expressed using the equation:

$$E = \frac{W}{Q}$$

- Where
  - E = electromotive force (e.m.f.), measured in volts (V)
  - W = energy transferred **to the charges** from the power source, measured in joules (J)
  - Q = charge moved, measured in coulombs (C)

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